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RESEARCH ARTICLE

Preparation, In Vivo and Molecular Docking Study of Nano-Emulsion Obtained From the Isolated Phytoconstituents of *Symplocos racemosa* for Mitigating Depression

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ABSTRACT

Depression is a prevalent, chronic, and debilitating disorder characterized by high treatment resistance. *Symplocos racemosa*, containing phenols and triterpenoids, acts as an antidepressant by elevating brain monoamine levels. This study aimed to enhance the targeted delivery of *S. racemosa* via nano-emulsion and improve its therapeutic efficacy against depression. In addition, it sought to elucidate the molecular mechanisms underlying its effects through molecular docking studies. *S. racemosa* bark was extracted using methanol and ethyl acetate, yielding six phytoconstituents (ellagic acid, betulinic acid, acetyl oleanolic acid, salireposide, oleanolic acid, and symlocoside) through column chromatography. Molecular docking against MAO-A was performed to confirm significant binding affinity. A chitosan-loaded herbal nano-emulsion containing these phytoconstituents was formulated for enhanced brain delivery tested using animal studies using forced swim test (FST) and actophotometer models. Docking studies confirmed that all the components exhibit a good binding toward MAO-A, comparable to standard moclobemide, and among all salireposide has the most significant and better binding affinity, higher than moclobemide. Nano-emulsion, at both dosages, reduced the immobility time in the FST and increased the locomotor activity in the actophotometer of depressed mice. Nano-emulsion of isolated phytoconstituents from *S. racemosa* exhibited synergistic effects, effectively managing depression by inhibiting MAO-A.

1 | Introduction

Depression is a common chronic and highly disabling disorder with a high level of treatment resistance. It is characterized by an overwhelming sense of despondency and a pervasive dampening of mental faculties and has emerged as the most prevalent mental health affliction in the general population, afflicting roughly 322 million individuals worldwide [1].

The majority of antidepressants in recent years have changed to *N*-methyl-D-aspartate (NMDA) receptor antagonists, which have a rapid onset of antidepressant effects. However, due to their adverse effects, which include hallucinations, addiction, cognitive impairment, and sensory disturbance, these medications are not employed as first-line antidepressants [2]. The main goal of treating the first symptoms of depression is to develop new therapeutic targets for antidepressants and medicinal products

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with short onset latency and fewer side effects. Because of their increased safety and fewer adverse reactions, herbal remedies play a significant role in disease prevention and treatment [3, 4] and are gaining more and more attention from scientists around the world.

Symplocos racemosa is identified as a diminutive evergreen tree reaching heights of 6–8.5 m [5]. Known as Lodhra in Sanskrit, denoting "auspicious," and Tilaka, as it was utilized for applying the forehead mark, this tree carries historical significance. Laden with a myriad of phytochemicals, the plant has garnered attention in traditional medicinal systems such as Ayurveda and Unani. The bark is extolled as an emmenagogue tonic suitable for individuals of the plethoric constitution, while its decoction serves as a gargle to fortify bleeding and spongy gums [6]. In addition, it addresses conditions associated with "Kapha," including biliousness, blood disorders, inflammations, vaginal discharges, leprosy, elephantiasis, and filaria. The traditional texts specify the inclusion of *S. racemosa* bark in 32 formulations, as documented in the Ayurvedic Pharmacopoeia of India (2001), highlighting its integral role in the treatment of diverse ailments [7]. A study demonstrated the rich existence of phenols such as ellagic acid, salireposide, and symplocoside and triterpenoids such as oleanolic acid, betulinic acid, and acetyl oleanolic acid (0.016 mg/g of dry weight of bark) [8].

Inculcating the herbal drug within a nanoformulation can increase their transport through the blood–brain barrier (BBB). Nano-emulsions (NE) are a kind of emulsion with droplet sizes ranging from 20 to 200 nm. These are used in an extensive spectrum of biological settings because of their tiny droplet sizes, which provide outstanding features, including durable stability and adjustable rheology. NEs are nano-sized emulsions made up of two insoluble components (water and oil) that are stabilized using one or two emulsifiers. NE is classified into the following categories: water-in-oil (W/O), oil-in-water (O/W), and oil–water–oil (O/W/O) [9]. Despite tremendous efforts in the domain of nano-based medication delivery, there have been limited achievements in developing a reasonable therapeutic treatment. Because of their capacity to carry drugs to the brain, NEs have the potential to administer antidepression therapies [10].

This study involved the isolation of six phytoconstituents from the plant *S. racemosa* and the utilization of a forced swim test (FST) and actophotometer on Swiss albino mice for in vivo evaluation of a chitosan-loaded herbal nano-emulsion (CLHN) prepared by mixing the isolated phytoconstituents. Furthermore, molecular docking studies were conducted to evaluate the binding affinities of these phytoconstituents within the active site of MAO-A.

2 | Materials and Methods

2.1 | Procurement of Plant

The botanical specimen was procured from the environs of Bhubaneswar, Odisha. The identification of the plant was accomplished through a meticulous examination of its distinctive features, including its foliage, fruits, and blossoms. R.K. Parmarthi, a distinguished researcher at the Indian Council of Agricultural Research-NBPGR, validated the authenticity of the complete botanical specimen. The identified plant was determined to be *S. racemosa* Roxb., with specimen No. AB-06/2024 a noteworthy member of the Symplocaceae family.

2.2 | Preparation of the Plant Extract

The extraction procedure was executed within an ultrasonic bath, utilizing a power of 560 W. A quantity of 10 g of *S. racemosa* coarse bark powder was measured and introduced into a conical flask filled with methanol. The ratio of solute to solvent employed was 10:50 (g/mL). The conical flask was then submerged in an ultrasonic bath, with temperature and duration parameters set at 30°C and 30 min, respectively. Following centrifugation at 990 rpm for 10 min, the resultant extract was separated from the solid residue. Subsequent concentration of the extract was achieved through the use of a rotary evaporator under vacuum conditions, operating at 40°C and 70 rpm [11]. The same process was repeated with the solvent ethyl acetate to yield an ethyl acetate extract.

2.3 | Isolation of Phytoconstituents From the Plant Extract

For isolating the phytoconstituents, two plant extracts (methanolic and ethyl acetate extract), as described in the above-stated step, were prepared.

Concentrated methanolic extract was subjected to fractionation by silica gel (60–120 mesh) column chromatography (CC) using a glass column (length: 43 cm, diameter: 3 cm). It was eluted with chloroform followed by a stepwise gradient of chloroform: ethyl acetate (100:0, 90:10, 80:20, 70:30 v/v) and ethyl acetate: methanol (100:0, 90:10, 80:20, 70:30 v/v) [12] to isolate ellagic acid. Similarly, to isolate acetyl oleanolic acid, the extract was acetylated [C_5H_5N–Ac_2O (1:1)] for 3 days and then eluted with petroleum ether: $CHCl_3$ (1:1 v/v) and subsequently mixed with 20 mL hexane and $CHCl_3$ (3:1 v/v) [13]. To isolate betulinic acid, the methanolic extract was further pre-adsorbed in a small portion of ethyl acetate, added to a small amount of silica gel, and dried before being put onto a slurry-packed silica gel column. The chromatographic column was systematically eluted using only *n*-hexane, followed by *n*-hexane:ethyl acetate mixtures as solvent systems (100:0, 90:10, 80:20, 70:30 v/v) [14]. For isolating salireposide, the methanolic extract was first extracted with ethyl acetate and then subjected to CC over a silica gel column. The column was initially eluted with hexane (100% v/v) followed by a stepwise gradient with hexane and $CHCl_3$ (90:10, 80:20, 70:30, 60:40, 50:50 v/v). This was followed by $CHCl_3$ (100% v/v), which helped extract moderately polar compounds. Subsequently, a $CHCl_3$:methanol gradient was used in ratios of 90:10, 80:20, and 70:30 v/v, leading to the final elution with methanol (100% v/v), which extracted the most polar compounds from the column [15].

To isolate oleanolic acid, the ethyl acetate fraction was subjected to CC on silica gel and eluted with different ratios of *n*-hexane and ethyl acetate (10:0, 9:1, 8.5:1.5, 8:2, 7:3, 5:5, 3:7, 2:8, 1:9, 0:10 v/v) [16]. In a similar manner to isolate symplocoside, the ethyl acetate fraction was subjected to silica gel CC and eluted with *n*-hexane (100% v/v), followed by a stepwise gradient of *n*-hexane

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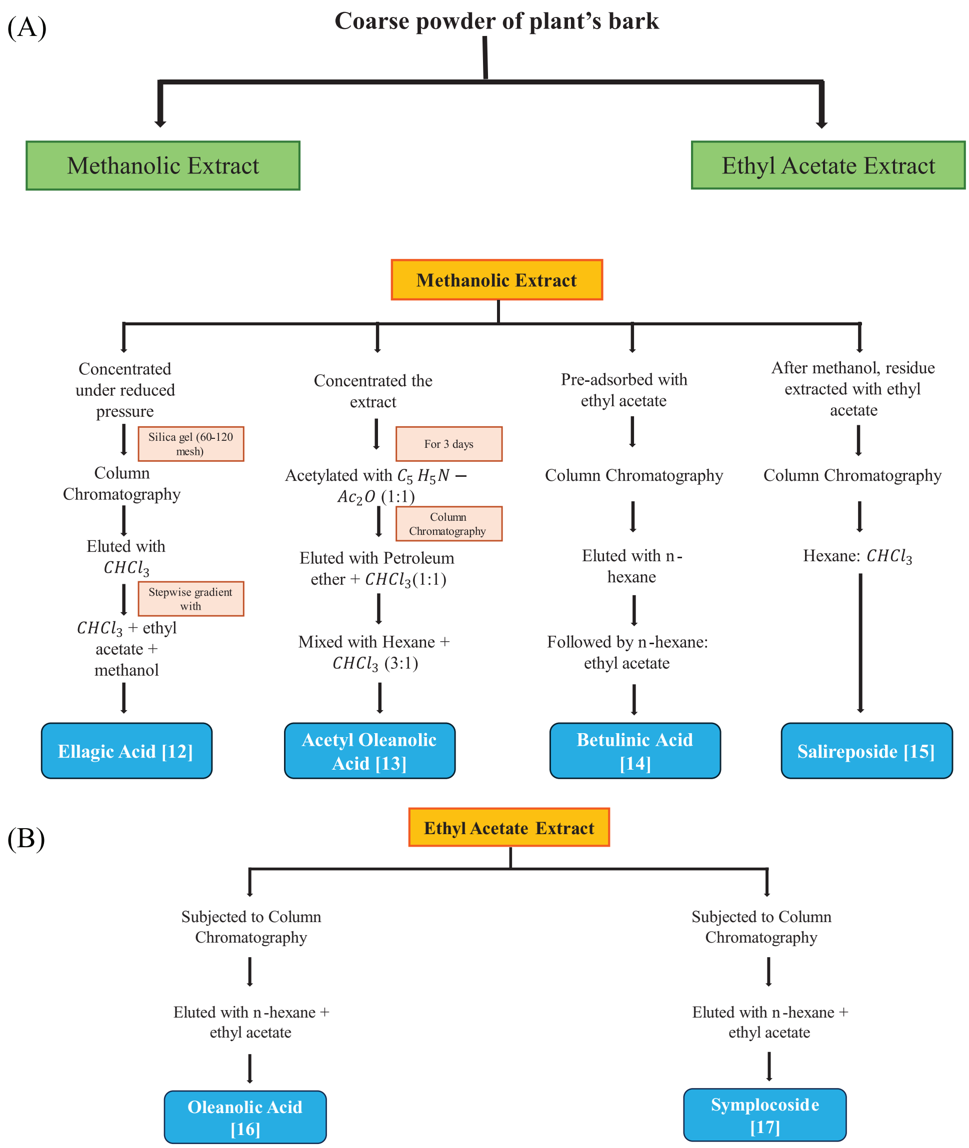

FIGURE 1 | (A). Isolation procedure from the methanolic fraction of *Symplocos racemosa*. (B) Isolation procedure from the ethyl acetate fraction of *S. racemosa*.

and ethyl acetate (90:10, 80:20, 70:30, 60:40, 50:50 v/v). This was followed by ethyl acetate (100% v/v) to elute more polar compounds. Finally, a gradient of ethyl acetate:methanol (90:10, 80:20, 70:30 v/v) was used, ending with methanol (100% v/v) to extract highly polar compounds [17]. The process of isolation is described in Figure 1A,B illustrated below.

2.4 | Molecular Docking

To understand molecular level interaction of phenolic (ellagic acid, salireposide, and symplocoside) and triterpenoid compounds (oleonalic acid, betulinic acid, and acetyl oleonalic acid) toward human monoamine oxidase A (MOA-A) (well-known therapeutic target for antidepressants), molecular docking studies were carried out using CB Dock-2, a web server which runs on a range of AutoDock Vina-based processes. The affinity of ligands for the active site of the enzyme is assessed by molecular docking. The crystal structure of ligands was obtained from the PubChem database, and the protein structure of MOA-A (PDB ID: 2Z5X) was downloaded from the Protein Data Bank. Protein was prepared by removing water molecules, merging nonpolar hydrogen atoms, and computing Gasteiger charge by assigning AD4 atom type. After preparation, docking was performed by CB Dock-2. All molecules in their respective largest clusters were further analyzed for their interaction with protein. Evaluating the number of distinct binding cavities in protein receptors and their interaction with ligands is the aim of in silico docking studies [18–20].

2.5 | CLHN

2.5.1 | Materials Required

Chitosan used as a rate-controlling polymer (Panvo Organics, Tamil Nadu, India), olive oil used as an oil phase (Mrinal Chemtech, New Delhi, India), PEG 400 used as a co-surfactant (Sigma-Aldrich, Bengaluru, India), and Span 20 and Tween 80

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used as surfactants (Indotec Chemicals & Polymers, New Delhi, India) of analytical grade were used for the preparation of CLHN.

2.5.2 | Synthesis of CLHN

A multiple O/W/O NE was formulated employing a two-step emulsification method [21].

2.5.2.1 | Primary O/W NE. The initial blending involved combining olive oil (29.5% w/w), a mixture of all the isolated constituents (ellagic acid, salireposide, symplocoside, oleonalic acid, betulinic acid, and acetyl oleonalic acid) (1% w/w), and Span 20 (1% w/w) in the oil phase, stirring for 15 min with a magnetic stirrer. Once the drug got completely dissolved in the oil phase, it was introduced to the aqueous phase, composed of distilled water (78.5% w/w), PEG 400 (4% w/w), chitosan (2% w/w), and Tween 80 (1% w/w). Subsequently, the amalgamation underwent homogenization at 10 000 rpm for 7 min using a high-shear mixer [21].

2.5.2.2 | Final O/W/O Multiple NE. The initial O/W NE (previously formulated) was introduced directly into the oil phase, comprising Span 20 (1% w/w) and olive oil (44% w/w), maintaining a proportion of 1:2 (NE to oil). This mixture was then homogenized for 7 min at 10 000 rpm using a high-shear blender [21].

2.5.3 | Characterization of CLHN

2.5.3.1 | pH of CLHN. A digital pH meter (New Era Scientific Corpn., Meerut, India) was used to measure the prepared preparation's pH level at ambient temperature. The average of the three measurements was computed as the result of repeating this process three times.

2.5.3.2 | Viscosity Measurement. Using a Brookfield viscometer (EIE Instruments Private Limited, Ahmedabad, India), the formulation's viscosity was measured without any dilution. At 30°C, spindle 62 assessed the thickness of the formulated mixtures. Rotating the spindle at 100 rpm for 1 min, the reading was recorded. This procedure was repeated three times.

2.5.3.3 | Cracking and Creaming. A 30 mL portion of prepared CLHN was extracted and placed into a glass container with a threaded lid. It was permitted to remain for one day at a temperature of 25 ± 2°C and then looked over visual traits. Cracking represents a form of physical instability, denoting the enduring/irreversible division or segregation of the dispersed phase (where water and oil exhibit distinct separation) located at the emulsion's upper part. The height of the whole emulsion and the height of the cream layer (upper layer) were calculated using the provided equation to estimate the creaming percentage in the case that the emulsions were separated into cream and serum components.

$$\%\ \text{Creaming} = \left(\frac{\text{Height of cream layer}}{\text{Total height of emulsion}}\right) \times 100$$

2.5.3.4 | Particle Size. A sample from the CLHN was placed into single-use sizing cuvettes and introduced into the measuring cell of the Litesizer 100 (Anton Paar India Private Limited, Gurugram, India) [21].

2.5.3.5 | Fourier Transform Infrared Analysis. Fourier transform infrared (FTIR) spectroscopy spectroscopic study of CLHN was performed to confirm the presence of functional groups corresponding to the six phytoconstituents by using IR Prestige 21, Shimadzu, Japan [22].

2.6 | Experimental Animal

For this study, male Swiss albino mice weighing from 25 to 30 g were utilized. The mice were procured from the Central Animal House at NIET (Pharmacy Institute) in Greater Noida. The experimental procedures involving animal subjects were sanctioned by the Institutional Animal Ethics Committee (IAEC) of NIET (Pharmacy Institute) under the reference number IAEC/NIET/2023/01/01. The mice were granted ad libitum access to water, ensuring uninterrupted availability, and were accommodated under regulated conditions with a temperature set at 22°C, relative humidity maintained between 50% and 70%, and a light–dark cycle of 12 h each.

2.7 | Acute Oral Toxicity Study of CLHN

The assessment of acute toxicity adhered to the protocols outlined by the Organization for Economic Cooperation and Development (OECD 423, 2001). Male Swiss albino mice were chosen as the subjects for this toxicity evaluation, with a total of three groups, each consisting of three mice. Before the commencement of the experimental protocol, the mice underwent an overnight fasting period. The dual objective of the acute toxicity study is to establish safe dose ranges for subsequent pharmacological investigations and observe any potential behavioral or pharmacological alterations arising from the administration of the CLHN. The orally administered CLHN was given to the mice at varying doses of 50, 300, and 2000 mg/kg body weight. Post-administration, the animals underwent continuous observation for the initial 4 h to identify any behavioral changes, and mortality assessments were conducted after 24 h, followed by daily observations spanning a 14-day period. The results indicated the CLHN's safety up to a single dose of 2000 mg/kg body weight. Consequently, the chosen doses for administration in experimental animals were established at 1/5th and 1/10th of the maximum safe dose, specifically at 400 and 200 mg/kg body weight [23].

2.8 | Drugs Used in Animal Experimentation

Moclobemide was acquired from Sigma-Aldrich in Bengaluru. All chemicals and solvents used during this study were of analytical quality.

2.9 | FST

2.9.1 | Grouping and Dosing of Animals

The animals were categorized into four cohorts, each having six animals. The treatment schedule is mentioned in Table 1.

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TABLE 1 | Animal experimental protocol.

Group	Treatment
Group I (control group)	Comprised mice that were administered with 10 mL/kg (po) of normal saline for 7 days
Group II (Standard group)	Mice received moclobemide at a dosage of 15 mg/kg (ip) for 7 days
Group III (Test Group 1)	Mice received 200 mg/kg (po) of CLHN for 7 days
Group IV (Test Group 2)	Mice received 400 mg/kg (po) of CLHN for 7 days

TABLE 2 | Animal experimental protocol.

Group	Treatment
Group I (Control group)	Comprised mice that were administered with 10 mL/kg (po) of normal saline for 7 days
Group II (Standard group)	Mice received moclobemide at a dosage of 15 mg/kg (ip) on the eighth day of experiment 1 h before subjecting to restraint stress
Group III (Test Group 1)	Mice received 200 mg/kg (po) of CLHN for 7 days, followed by exposure to acute restraint stress on the eighth day
Group IV (Test Group 2)	Mice received 400 mg/kg (po) of CLHN for 7 days, followed by exposure to acute restraint stress on the eighth day

2.9.2 | Procedure

Mice were subjected to tests and standard drug treatments as per the above-mentioned protocol. On the seventh day, following an hour of administering medication, animals were evaluated for antidepressant effects. This test comprises a single session lasting for 6 min, divided into a pretest phase (initial 2 min) and a test phase (final 4 min). The experimental protocol required mice to navigate a cylindrical enclosure measuring 15 cm in diameter and 25 cm in height, undergoing this task for a duration of 6 min. The cylinder contained 15 cm of freshwater maintained at $25 \pm 1°C$, and the depth of the water was tailored according to the mouse's dimensions to prevent contact between its hind limbs and the container's base. After every animal, the water in the reservoir was changed to avoid alterations in animal behavior brought on by sharing the same water. After a duration of 6 min had passed, the mouse was taken out of the container, dried with a cotton towel, and positioned in a temporary dry cage featuring an overhead heat lamp and a lower heat pad. The mice were subjected to rigorous and constant scrutiny during their recuperation within this cage. During the initial 2-min phase of the test, all animals exhibited vigorous locomotor activity. Subsequently, the duration of inactivity was carefully noted for the remaining 4 min out of the complete 6-min testing period. Mice were classified as immobile when observed in a vertical flotation posture, exhibiting minimal activity primarily aimed at sustaining their head above the water surface. A decrease in the period of immobility indicates a potential antidepressant effect, whereas an elevation in immobility duration relative to the animals of the control group is associated with depressive-like manifestations [24].

2.10 | Actophotometer Model

2.10.1 | Grouping and Dosing of Animals

The animals were categorized into five cohorts, each having six animals. The treatment schedule is mentioned in Table 2.

2.10.2 | Procedure

To assess locomotor activity, an actophotometer was utilized. This instrument functions by detecting the interruption of a light beam resulting from the motion of the animals and subsequently capturing and digitally presenting the recorded data. Swiss albino mice received 200 and 400 mg/kg of CLHN for a duration of 7 days. On the eighth day of the experiment, before subjecting to restraint stress, moclobemide, a typical antidepressant medication, was administered via intraperitoneal injection at a dosage of 15 mg/kg. All mice, except for those in the control group, underwent exposure to acute restraint stress (ARS).

Stress through immobilization was triggered by placing each mouse in separate rodent restraint devices made of wire mesh for a period of 6 h. This limited all bodily movements but did not inflict pain on the mice. During the entire period of stress exposure, the mice had no access to either food or water. After the 6-h confinement, the mice were freed from their restraints, and following a 40-min waiting period, their movement activity was evaluated using an actophotometer. In the control cohort, mice were housed conventionally in standard cages situated within the experimental chamber. Mice were positioned in an actophotometer, and the quantity of motions inside the device was tallied over a 5-min period. The equipment was purified using a 10% ethanol mixture after each trial to conceal animal traces [25].

2.11 | Statistical Analysis

The readings were reported using both mean ± standard deviation (SD) and mean ± standard error mean (SEM). Group comparisons were performed through ANOVA, supplemented by subsequent application of Tukey's honestly significant difference (HSD) test for detailed pairwise comparisons. Statistical

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TABLE 3 | Molecular interaction of MAO-A active site with phenolic and triterpenoid compounds.

Category	Ligand	MAO-A		
		Docking score (kcal/mol)	No. of H-bonds	H-bond interacting residues
Phenols	Salireposide	**−10.4**	4	Tyr69, Gly443, Ile180, Asn181
	Ellagic acid	−8.9	3	Tyr444, Tyr197, Gly443
	Symplocoside	−9.0	3	Glu185, Asp328
Triterpenoids	Oleanolic acid	−8.7	1	Asn179
	Betulinic acid	−8.0	1	Thr205
	Acetyl oleanolic acid	−8.0	4	Arg109, Phe112, Val210, Thr211
Standard	Moclobemide	**−8.9**	3	Tyr402, Ala44, Arg51

Note: The bold values indicate the docking scores of the standard drug (moclobemide) and the isolated phytochemical salireposide. These values highlight the comparison, showing the docking score of the phytoconstituent which is closest to the standard drug.

significance was indicated by values of $p < 0.05$, $p < 0.01$, $p < 0.001$, and $p < 0.0001$, respectively.

3 | Results

3.1 | Yield Analysis of *S. racemosa*

The dehydrated bark of *S. racemosa* was subjected to extraction using methanol and ethyl acetate as a solvent, employing the ultrasonic bath. The resultant plant extract was subsequently desiccated using a heated water bath, and the percentage yield for the plant was determined to be 19.2% w/w with methanol and 20.04% w/w with ethyl acetate.

3.2 | Isolation

Using methanolic and ethyl acetate extracts of *S. racemosa*, six different phytoconstituents, namely ellagic acid, betulinic acid, acetyl oleanolic acid, salireposide, oleanolic acid, and symlocoside, were isolated using CC.

3.3 | Molecular Docking Studies

The free energy of binding with most clusters in the clustering histogram is interpreted as its docking score. The docking score (vina score) of the phenols and triterpenoids with the target protein (MAO-A) is in a range from −8 to −10.4 kcal/mol, which was compared to the standard drug moclobemide having −8.9 kcal/mol with MAO-A and binding with the various amino acid residues with the heterocyclic rings and the functional groups. A detailed description of ligands with their respective targets and amino acid residues interacting with the ligands is mentioned in Table 3. An in silico study found that salireposide contains a strong binding affinity with MAO-A (−10.4 kcal/mol) along with strong molecular interaction by forming several hydrogen bonds, which were better than the standard drug moclobemide. Salireposide showed interaction with amino acid residues Tyr69, Gly443, Ile180, and Asn181 at the active site of MAO-A (Table 3). Hydrogen bonds stabilize the ligand at the active site. 2D and 3D molecular docking interactions of all the ligands with MAO-A are depicted in Figure 2.

3.4 | Characterization of the CLHN

3.4.1 | pH of CLHN

Following the pH measurement of the CLHN with a digital pH meter, it was discovered that the NE's pH was 7.9 ± 0.02, deemed suitable for oral delivery.

3.4.2 | Viscosity Measurement

The fundamental characteristic of a NE is its viscosity, which is influenced by water, oil, and surfactant concentrations. Elevating the water content in an NE results in decreased viscosity. This property is crucial for NEs as it affects stability and drug release patterns. By utilizing a Brookfield viscometer spindle 62, the viscosity of CLHN was determined to be 60 cP, falling within the optimal viscosity range (10–100 cP) for oral nanosuspension.

3.4.3 | Creaming and Cracking

Following 1-day storage of the CLHN, visual examination showed no evidence of cracking, signifying the absence of phase division. This observation suggests the physical stability of the prepared herbal multiple NE.

3.4.4 | Particle Size Measurement

CLHN exhibited a particle size of 198.55 nm. The graph of particle size obtained via Litesizer 100 is demonstrated in Figure 3.

3.4.5 | FTIR Analysis

The FTIR profile of CLHN displayed distinctive absorption peaks, verifying the inclusion of the six bioactive compounds along

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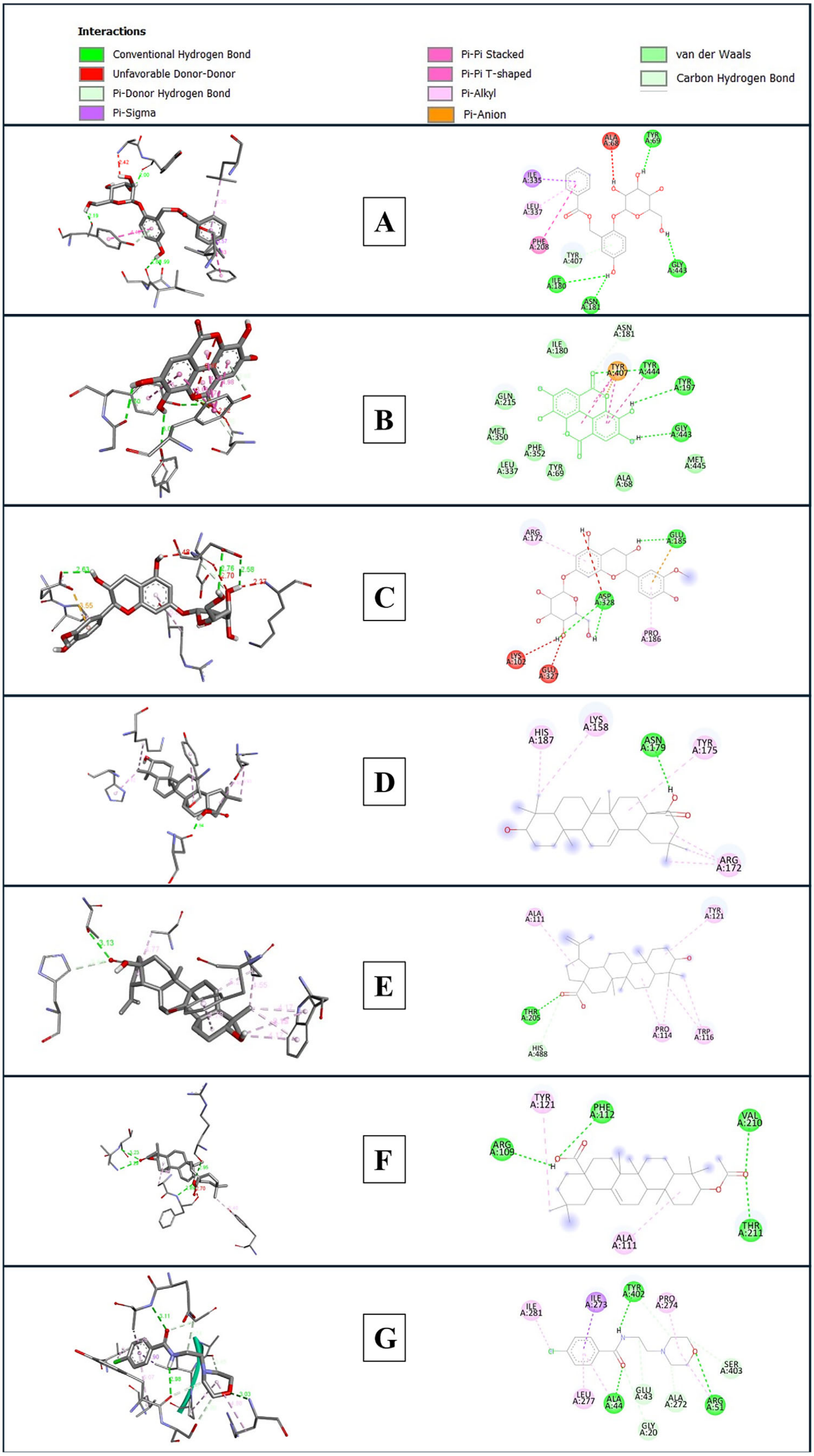

FIGURE 2 | 2D and 3D molecular docking interactions of (A) salireposide, (B) ellagic acid, (C) symplocoside, (D) oleanolic acid, (E) betulinic acid, (F) acetyl oleanolic acid, and (G) moclobemide with MAO-A protein.

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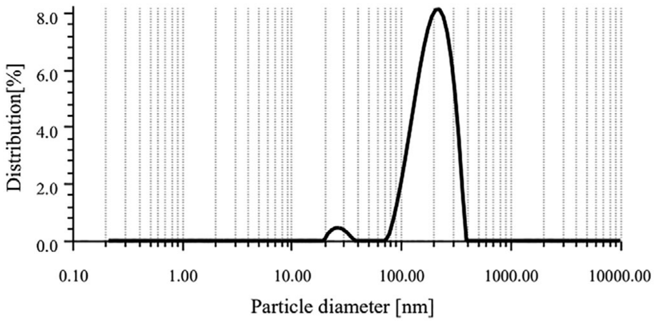

FIGURE 3 | Particle size measured by Litesizer 100.

with chitosan. Figure 4 presents the FTIR spectrum of CLHN, highlighting the presence of distinct functional groups that correspond to the phytochemicals integrated into the formulation. The absorption bands at 1653.23, 1684.77, and 1717.75 cm^{-1} correspond to the carbonyl (C═O) stretching vibrations of carboxyl and ester functional groups, affirming the presence of oleanolic acid, betulinic acid, and acetyl oleanolic acid. Furthermore, the 1653.23 cm^{-1} peak aligns with amide I (C═O) stretching, signifying the integration of chitosan in the formulation. The absorption peak at 1560.03 cm^{-1} falls within the amide II (N—H bending) region, further corroborating the incorporation of chitosan within CLHN. The bands at 1506.98 and 1560.03 cm^{-1} can be attributed to C═C stretching in aromatic rings, indicative of ellagic acid and symplocoside. Peaks identified at 1028.07 and 1319.14 cm^{-1} correspond to C—O stretching, characteristic of glycosidic linkages found in salireposide and symplocoside, along with C—O—C bonds of chitosan. In addition, the absorption bands at 668.17 and 778.58 cm^{-1} are assigned to C—H bending vibrations of aromatic structures, reinforcing the presence of phenolic compounds. These observations confirm that the primary phytochemicals are effectively incorporated into CLHN without any substantial structural modifications, ensuring its chemical stability and formulation integrity.

3.5 | Acute Toxicity Study

In the realm of pharmaceutical exploration and advancement, the evaluation of drug toxicity profiles holds paramount significance in assessing their effectiveness against targeted ailments. Within this study, the acute oral toxicity profiles of the CLHN were meticulously scrutinized, following the established protocols outlined by the OECD 423. These guidelines follow a stepwise procedure that minimizes the use of animals at each stage. The evaluation of acute oral toxicity for the CLHN was conducted in accordance with these guidelines.

The acute toxicity investigation outlined in the aforementioned Table 4 reveals a conspicuous absence of fatalities across all examined concentrations, spanning from 50 to 2000 mg/kg. Vigilant monitoring of the test subjects immediately after dosing was carried out, with continuous scrutiny of their behavior within the initial 4 h and evaluation for mortality after 24 h, followed by daily observation for a duration of 14 days. These meticulous observations validate the safety of the prepared CLHN at a singular dosage of 2000 mg/kg body weight. Consequently, the dose of 400 and 200 mg/kg b.w was judiciously selected for administration in the experimental animals, representing 1/5th and 1/10th of the maximum permissible dosage, respectively.

TABLE 4 | Acute toxicity study of CLHN.

Group	Dose (mg/kg) orally	Mortality	
		24 h	**48 h**
CLHN	50	No	No
	300	No	No
	2000	No	No

3.6 | FST

On the seventh day of the test, the control group showed an average immobility time of 4.20 ± 0.39 s, which serves as a baseline for comparison with other treatments. Animals treated with the standard drug moclobemide exhibited a significant reduction in immobility time of 1.44 ± 0.13 s ($p < 0.0001$) compared to the control group. Administration of CLHN at a dosage of 200 and 400 mg/kg exhibited significant reduction ($p < 0.0001$) in immobility time 1.55 ± 0.47 and 1.35 ± 0.59 s (Figure 5).

3.7 | Antidepressive Activity by Locomotor Activity (Actophotometer Model)

Acute treatment with CLHN for 7 days and subjecting mice to ARS on the eighth day as a source to induce depression significantly increased the activity score with 200 mg/kg (310.5 ± 5.23; $p < 0.0001$) and 400 mg/kg (317.83 ± 7.67; $p < 0.001$) in contrast to the control group (278.0 ± 1.58), which is a characteristic of good locomotor activity denoting an antidepressant-like effect. However, those animals who received the standard drug moclobemide showed the highest significant increment in activity score (338.5 ± 5.14; $p < 0.001$) (Figure 6).

4 | Discussion

S. racemosa, according to literature, has an abundance of triterpenoids, including acetyl oleanolic acid (0.016 mg/g of dry weight of bark), oleanolic acid, and betulinic acid, as well as phenols such as ellagic acid, salireposide, and symplocoside, which may be in charge of the neuroprotective properties of *S. racemosa* [8]. With the help of CC, we isolated all these selected six components from *S. racemosa* and performed their molecular docking.

A potential method for improving the effectiveness and administration of antidepressant medications is the use of oral NEs. The usual droplet size of NEs is between 20 and 200 nm. Their compact size and special qualities can increase therapeutic drugs' solubility, stability, and bioavailability, enabling more efficient treatment of depression [26]. When used in oral NEs, the natural polysaccharide chitosan has attracted a lot of interest due to its potential to improve medication delivery across the BBB [27]. Our formulated NE demonstrated optimal pH, particle size, viscosity,

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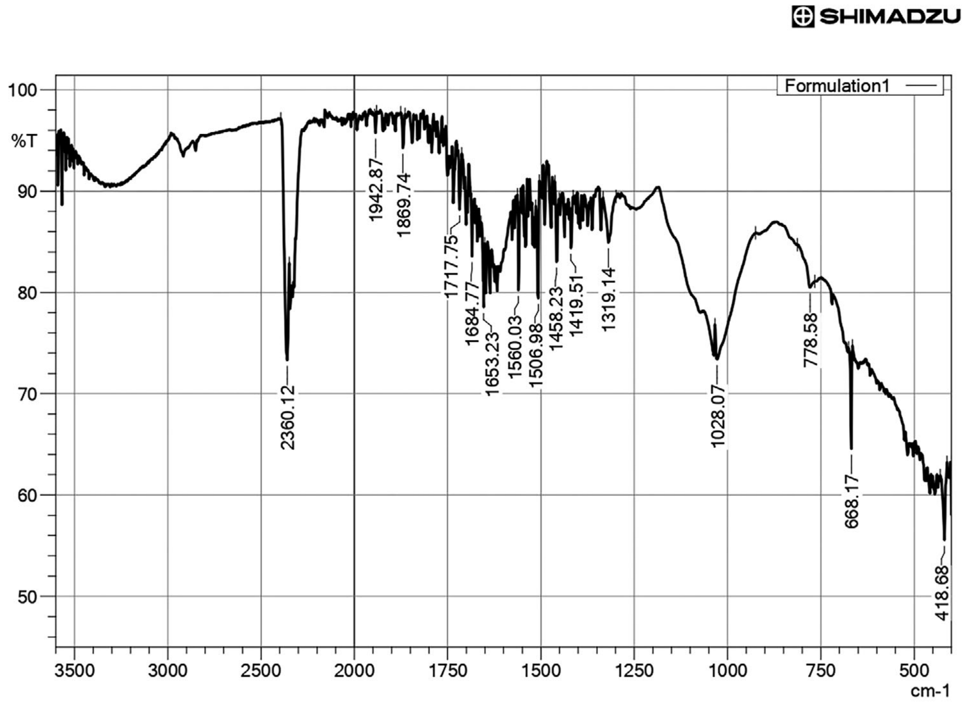

FIGURE 4 | FTIR spectrum of CHLN.

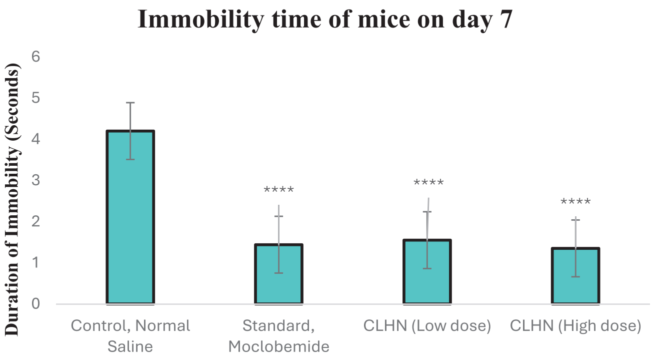

FIGURE 5 | Effect of CLHN on duration of immobility using FST. n = 6, all values represent mean ± SD; ****$p < 0.0001$ (one-way ANOVA followed by post hoc Tukey HSD).

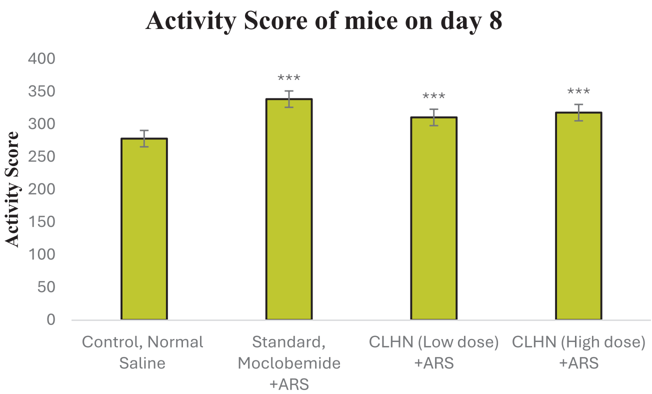

FIGURE 6 | Effect of CHLN on locomotor activity using actophotometer model. n = 6, all values represent mean ± SEM; *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$ (one-way ANOVA followed by post hoc Tukey HSD).

and physical stability, making it suitable for oral delivery. FTIR analysis further confirmed the successful incorporation of key phytoconstituents, ensuring the structural integrity and stability of CLHN.

Although the exact cause of depression is still unknown, HPA axis disruption has been found to be a significant risk factor [28]. The abnormal regulation of the negative feedback on glucocorticoid amounts, including cortisol, is the outcome of HPA axis malfunction. As seen in many depressed people, this results in elevated cortisol levels [29]. Increased cortisol levels cause astrocytes in the CNS to become more MAO active [30]. During this process, the synthesis of neurotransmitters, including dopamine and serotonin in the CNS, is inhibited by MAO-induced ROS formation [31]. The depression-like behavior in FST was induced by placing mice in water and by restraining all movements of mice (immobilization stress) in the actophotometer model, which effectively resembles HPA axis dysfunction. Control animals in both models showcased increased immobility time in the FST and reduced movement (activity score) in the actophotometer, mimicking disease phenotypes in humans. These behaviors can be significantly improved by treatment with antidepressant drugs or natural products [32]. Similarly, our results showed that control mice exhibited increased immobility in the FST and reduced activity scores in the actophotometer. However, the CLHN-treated group revealed a marked decrease in immobility time with 200 and 400 mg/kg (1.55 ± 0.47 and 1.35 ± 0.59 s; $p < 0.0001$), indicating potential antidepressant effects comparable to the standard drug moclobemide. The actophotometer model supported these findings by showing increased locomotor activity with 200 mg/kg (310.5 ± 5.23; $p < 0.0001$) and 400 mg/kg (317.83 ± 7.67; $p < 0.001$) in the test groups, suggesting reduced central nervous system drowsiness. These in vivo findings were supported by the results of molecular docking. It is a practical

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computational technique for forecasting how tiny compounds will interact with proteins in molecular docking. It makes it easier to recognize possible binding modes and comprehend the underlying mechanics [33]. Strong MAO-A inhibitors have been created from natural compounds using molecular docking analysis [34]. Our molecular docking studies revealed that all six components isolated via CC interacted with MAO-A by strongly binding to its active site to the same degree as moclobemide, the standard inhibitor. Among all, salireposide showed the highest binding with MAO-A. Thus, we postulated that all six isolated components present in CHLN show a synergistic activity, with salireoside being the main active component (due to the highest dock score) of *S. racemosa*, leading to MAO-A inhibition. These findings indicate that CLHN possesses antidepressant properties (via inhibition of MAO-A) in animal models of depression.

5 | Conclusion

In this study, we prepared methanolic and ethyl acetate extracts of *S. racemosa* and isolated six phytoconstituents from the same. Then we conducted molecular docking of these six phytoconstituents with the antidepressant target MOA-A. Moving further, we formulated a CLHN with the mixture of isolated phytoconstituents and tested this formulation through in vivo antidepressant models, namely, the FST and actophotometer.

The prepared CLHN, at higher efficiency, exhibited reduced immobility time in FST and increased locomotor activity of mice in the actophotometer, indicating its antidepressant potential. These in vivo results were supported through in silico molecular docking, which showed that all six isolated phytoconstituents had good binding with MOA-A, and among all, the present salireposide in *S. racemosa* bark showed the highest dock score, which could be the main constituent for showcasing the antidepressant effect. This highlights that the mixture of six phytoconstituents could have worked synergistically to show an antidepressant effect (via inhibition of MAO-A). Further studies can be conducted to explore the therapeutic potential of this plant, as well as work on various other formulations for better-targeted delivery, reaching the brain.

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Ethics Statement

The study has been approved by the Institutional Animal Ethics Committee of Noida Institute of Engineering and Technology (Pharmacy Institute), India, under the reference number IAEC/NIET/2023/01/01. The male Swiss albino mice (25–30 g) were obtained from the Central Animal House of NIET (Pharmacy Institute) in Greater Noida. The mice were given water ad libitum, providing constant access, and were kept in regulated settings with a temperature of 22°C, relative humidity that ranged from 50%–70%, and a 12:12 light–dark cycle.

Consent

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The authors have nothing to report.

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